

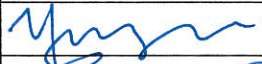
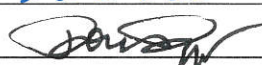
		CONTROLLED FORM		
TITLE: GXP and Non-GXP Study Report Review and Approval Form				
DOCUMENT NO. GE003FM01	VERSION 02	ISSUE DATE 17 Oct 2014	EFFECTIVE DATE 20 Oct 2014	PAGE 2 of 2

☐ Vendor Study Protocol ☒ Vendor Study Report ☐ Lpath Study Report					
Lpath Study No.: LT3114-SPH-001-R		Version No.: 01		☒ Non GXP ☐ GCP	☐ GLP ☐ GMP
Product Name: LT3114		Test Article: LT3114			
Study Title: A study to assess wound healing in C57 BL/6 mice					
Study Vendor/Site: ☐ N/A Covance, Greenfield IN		Study Director: ☐ N/A Rosalia Matteo		Study Contributor(s): ☐ N/A Ying Li	
Vendor Study No./Version: 8300180		Study Dates: July 2014		Report Date: December 2014	
Lab Notebook Reference: LNB# 316 pp. 44-47; 65-72 ☐ N/A					

Executive summary of study results: ☐ See Attached ☐ N/A

The purpose of this study was to test the effects of test article LT3114 at two doses, in a mouse model of wound healing by evaluating wound closure and tensile strength. In this study, C57 BL/6 mice (n=8/group) were given twice weekly subcutaneous doses of 0, 10, or 50 mg/kg for 2 weeks.

Animals treated with LT3114 at 10 mg/kg showed a significant effect on the wound healing process in terms of accelerated wound closure compared with vehicle treated animals but demonstrated no change in tensile strength of the healed wound. Animals treated with LT3114 at 50 mg/kg also exhibited significant improvement of wound closure compared with vehicle treated animals but no change in tensile strength.

LPATH APPROVAL			
Function	Name (Print)	Signature	Date
Study Director/Monitor	Rosalia Matteo		12/08/2014
Contributor	YING LI		12/8/14
Research	GARY WOODWORTH		8th December 2014
Development	DARIO PABBIANO		DEC 12, 2014
Quality Assurance	ELVIRA MARQUEZ		12 DEC 14

Amended Final Report

Study Title	A Study to Assess Wound Healing in C57 BL/6 Mice
Test Article	LT3114
Study Director	Tricia Wolff
Testing Facility	Covance Laboratories Inc. 671 South Meridian Road Greenfield, Indiana 46140 United States of America
Sponsor	Lpath 4025 Sorrento Valley Blvd San Diego, California 92121 United States of America
Covance Study Number	8300180
Report Issued	08 December 2014
Amended Report Issued	30 December 2014
Page Number	1 of 41

SIGNATURE PAGE

The following Covance personnel are contributing scientists for their respective Results and Discussion sections and agree these sections contain their interpretation of these data.

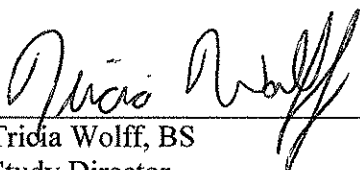


Subba R. Chintalacharuvu, PhD
Scientific Lead
Covance Laboratories Inc.

12-30-14

Date

The following individual is responsible for the overall conduct of this nonclinical laboratory study and for the data reported herein.



Tricia Wolff, BS
Study Director
Covance Laboratories Inc.

30-Dec-2014

Date

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SUMMARY AND CONCLUSIONS

The purpose of this study was to test the effects of test article LT3114 at two doses, in a mouse model of wound healing by evaluating wound closure and tensile strength. In this study, C57 BL/6 mice (n=8/group) were given twice weekly subcutaneous doses of 0, 10, or 50 mg/kg for 2 weeks.

Animals treated with LT3114 at 10 mg/kg showed a significant effect on the wound healing process in terms of accelerated wound closure compared with vehicle treated animals but demonstrated no change in tensile strength of the healed wound. Animals treated with LT3114 at 50 mg/kg also exhibited significant improvement of wound closure compared with vehicle treated animals but no change in tensile strength.

Concentrations of LT3114 in plasma collected at the end of study were assayed by both human IgG quantification or by binding ELISA. All animals had quantifiable concentrations in the plasma and in general the two methods of quantification were similar. However there was no dose response noted for LT3114 between 10 to 50 mg/kg.

Wound healing is a highly organized process involving various cell types. The wound healing process can be divided into three distinct but overlapping phases: 1) hemostasis and inflammation; 2) proliferation and 3) maturation or remodeling. Based on the results, LT3114 at 10 mg/kg seems to exhibit an effect on the inflammation phase based on the percent wound closure, whereas it had no effect on the maturation or remodeling phase as measured by tensile strength. However, effects of LT3114 on inflammation need to be further confirmed by histological evaluation (e.g. dermal gap and dermal hyperplasia).

STUDY INFORMATION AND PROCEDURES**Key Personnel**

Study Director	Tricia Wolff
Scientific Lead	Subba Chintalacharuvu
Sponsor Representative	Gary Woodnutt, Lpath Inc.

Study Dates

Study Start Date	16-Jul-2014 (Day 1, wound surgery)
Study End Date	29-Jul-2014 (Day 14; scheduled necropsy)

Test and Vehicle Control Articles

Test Article	LT3114
Lot Number, Concentration	GA121459, 32.1 mg/mL
Vehicle	Formulation Buffer

Test Animal

Species, Strain	Mouse, C57BL/6
Number of Animals	8 females/group
Age	10 to 11 weeks at study start

Surgical Procedure

Pre-Surgery Preparation	One day prior to surgery, all animals were shaved and a depilatory (Nair®) was applied to the surgical area to remove the hair. On the day of surgery the animals were anesthetized to effect using 1% to 5% Isoflurane gas in oxygen. Eye lubricant was placed in the animals eyes to prevent drying of the corneas during anesthesia. The mice were dosed with 0.05 mg/kg Buprenorphine IM/SC. The skin at the wound sites was scrubbed with appropriate antiseptic (i.e. betadine, chlorhexidine) and wiped with sterile saline or isopropyl alcohol solution.
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Surgery	On Day 1 two wounds were made as follows: <u>Cranial wound:</u> On the level of the dorsal midline at the scapula/thoracic area, a sterile 6-mm punch biopsy tool was used to cut through the epidermis and dermis, and then scissors were used to detach the tissue. Full thickness wounds extending through the panniculus carnosus were made using the scissors. <u>Caudal wound:</u> Approximately 1 cm caudal to the cranial wound, a 1-cm linear incision was made through the skin and subcutis along the dorsal midline. After photography, 20 μL of saline was administered directly to the wound. Tincture Benzoin compound was applied outside of the perimeter of the wounds and the semi permeable polyurethane dressing OpSite will be placed over the wound and sealed at the edges by the Benzoin.
Post-Operative Care	Following the procedures, the animals were transferred to a warm recovery cage and monitored until fully recovered. Animals were then returned to their home cage. Signs of pain, swelling, infection or other abnormalities were monitored.

Dosing Regimen

Route	Subcutaneous injection		
Dosing Days	1, 4, 8, and 11		
Dose Groups	Dose (mg/kg)	Dose Volume (mL/kg)	Animals
	0 (vehicle control article)	5	Group 01
	10	1	Group 02
	50	5	Group 03

Study Parameters

Pharmacokinetics

Sample Collection	Blood was collected via cardiac puncture from all animals at necropsy. Samples were collected into tubes containing K ₃ -EDTA and processed to plasma.
Analysis	Samples were shipped to the sponsor for possible future analysis.

Live Phase

Clinical Observations	On dosing days, clinical observations were recorded daily before dosing and approximately 1 hour postdose.
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Body Weights	Animals were weighed prior to surgery and on Days 7 and 13.
Photography	Photographs of the wounds were taken on Days 1, 3, 6, 9, 13, and terminal. The wound diameter was determined using Image J.

Anatomic Pathology

Tissue Collection	A rectangular section of skin was removed, including the subcutis that contained the wound and extended approximately 2 cm lateral to the wound edges. The 2 wound sites were separated. The cranial wound was assessed for wound tensile strength as described below. Due to the progression of wound healing the caudal wound was discarded without further evaluation.
Wound Tensile Strength	The skin section was collected into 0.9% saline and positioned individually in the tensiometer grips. The tensiometer then applied stretching force at a fixed rate to the skin until sample failure (the point the skin is torn). The instrument reported the maximum force that caused the sample to failure. After failure, the remaining pieces were discarded.

Statistical and Data Analysis

Wound Closure and Wound Tensile Strength	One-factor ANOVA with trend and Dunnett's (if applicable) and percent change from control
AUC Calculation	The AUC was calculated using the trapezoidal rule, with percent change in wound closure on Y-axis and days post injury on X-axis. The AUC is expressed as % wound closure*days and represents total wound healing over time.

Protocol Deviations

-
- On Day 4, 19-Jul-20114, the 1 hour post dose observations were performed approximately 2 hours post dose.
 - The protocol deviation described neither affected the overall interpretation of study findings nor compromised the integrity of the study.
-

RESULTS AND DISCUSSION

Live Phase

Survival and Clinical Observations

All animals survived to study termination. No compound-related changes in clinical condition occurred. A few animals were noted with skin abrasions on the dorsal region. These abrasions were not considered important because they were only noted in a few animals across all groups.

Body Weights

No important changes in body weight parameters occurred. All groups were recorded

Plasma concentrations of LT3114

Concentrations of LT3114 in plasma collected at the end of study were assayed by both human IgG quantification or by binding ELISA (see Pharmacokinetic Report). All animals had quantifiable concentrations in the plasma and in general the two methods of quantification were similar. However there was no dose response noted for LT3114 between 10 to 50 mg/kg.

Wound Size Closure and Tensile Strength

Two types of wounds including the cranial wound (6 mm punch biopsy) and the caudal wound (1 cm linear incision) were created in each animal. The cranial wound was evaluated for percent wound closure and tensile strength measurement at the terminal stage of the experiment. The initial wound size was measured before the administration of vehicle or LT3114 at 10 mg/kg or compound 50 mg/kg (n=8/group). The changes of wound size were measured on Days 3, 6, 9, 13 and 14 post wound creation.

Three days post wound creation, all groups exhibited an increase in wound size but closure of wound was observed on Day 6 for animals treated with compound LT3114 treatment whereas the vehicle only showed a closure response starting on Day 9. On Days 9, 13, and 14, the mean $\pm$ SE percent wound closure in vehicle treated animals was 22.21 % $\pm$ 9.06%, 82.45% $\pm$ 4.94% and 97.65%, respectively. Animals treated with LT3114 at 10 mg/kg exhibited 24.81% $\pm$ 5.02%, 49.36 $\pm$ 2.92%, 92.77 $\pm$ 1.61%, 99% $\pm$ 0.69% wound closure on Days 6, 9, 13 and 14 respectively (Figure 1A). Animals treated with LT3114 at 10 mg/kg healed significantly faster (* p < 0.0006; Dunnett-t test) compared with vehicle treated animals (Figure 1B). At 50 mg/kg of LT3114, the percentage of wound closure was 5.69% $\pm$ 11.04%, 46.78% $\pm$ 5.16%, 94.76% $\pm$ 1.52%, 99.62% $\pm$ 0.13% on Days 6, 9, 13 and 14, respectively. Animals treated with LT3114 at 50 mg/kg also demonstrated significantly faster wound healing (* p < 0.0101; Dunnett-t test) compared with vehicle treated animals. Finally, animals treated with LT3114 at two different doses (i.e. 10 mg/kg and 50 mg/kg) showed a similar trend in healing (p < 0.2526; Dunnett-test test).

Breaking strength was evaluated at the end of the experiment using the cranial wound. The tensile strength was in the normal range (between 9.37 g/mm² to 10.86 g/mm²) and

there was no difference among the three groups (Figure 2).

In conclusion, animals treated with LT3114 at 10 mg/kg showed a significant effect on the wound healing process in terms of accelerated wound closure compared with vehicle treated animals but demonstrated no change in tensile strength of the healed wound. Animals treated with LT3114 at 50 mg/kg also exhibited significant improvement of wound closure compared with vehicle treated animals but no change in tensile strength.

FIGURE

Figure 1

A. Time course of wound closure in C57BL/6 mice (n=8/group). On day 1, 4, 8 and 11, animals were dosed subcutaneously with vehicle or two different doses of LT3114 (10 mg/kg and 50 mg/kg). Wound area was measured immediately after wound creation and at various times during the healing process. Values are presented as percentage of original area (means $\pm$ SE).

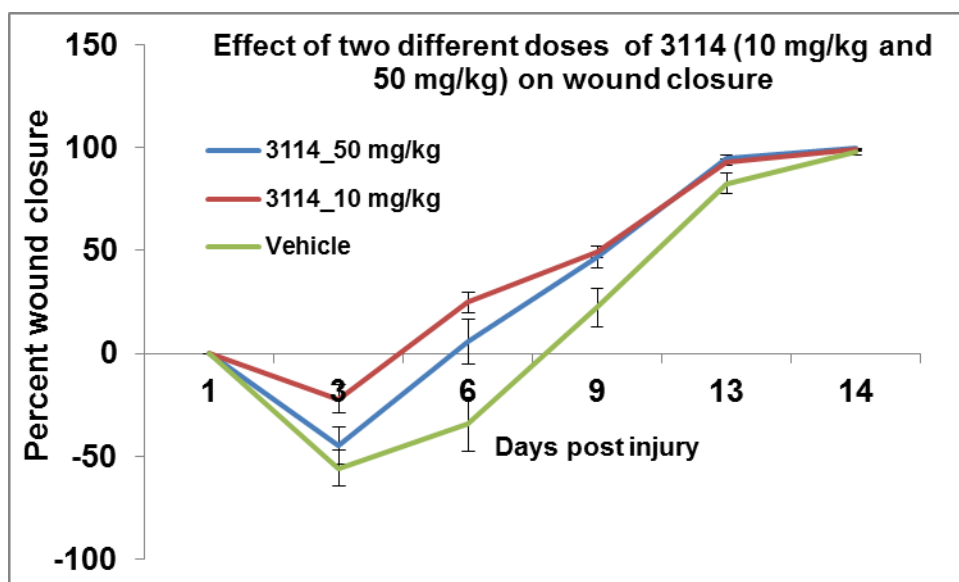


Figure 1 continued

B. Area under curve (AUC; percent closure/day) was calculated based on the percent wound closure measured on day 1, 3, 6, 9, 13 and 14 post-injury. On day 1, 4, 8 and 11, animals were dosed subcutaneously with vehicle or two different doses of LT3114 (10 mg/kg and 50 mg/kg; n=8/group). Values are presented as AUC (means $\pm$ SE).

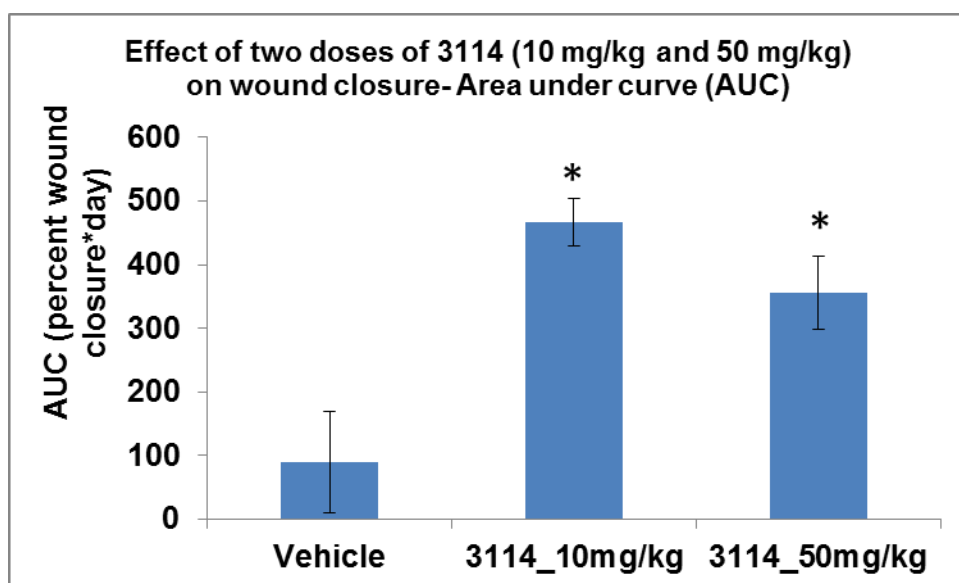
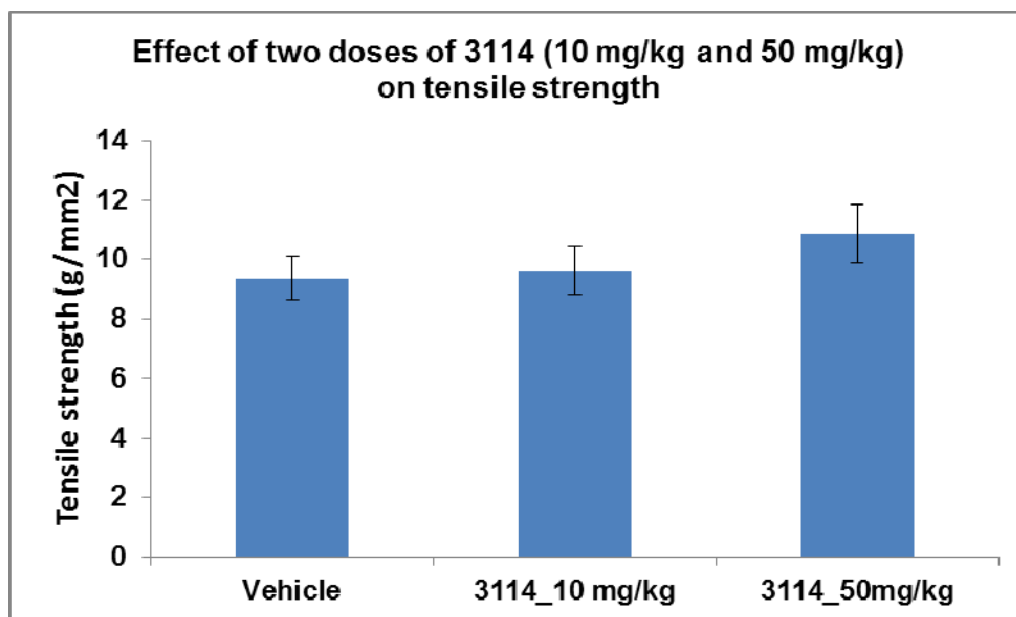


Figure 1

Tensile strength measured on the terminal day i.e. day 14 post-injury in C57BL/6 mice (n=8/group). On day 1, 4, 8 and 11, animals were dosed subcutaneously with vehicle or two different doses of LT3114(10 mg/kg and 50 mg/kg). Values are presented as tensile strength (means $\pm$ SE).



TABLES

Table 1
Summary of Clinical Observations

Clinical Signs Summary Report by Days

Page 1 of 1

Covance

Printed: 16 Sep 2014 11:08:50 AM

PATH/TOX SYSTEM Version 1.6.0 Build 27

Study: 8300180

StudyTitle: A Study to Assess Wound Healing in the C57BL/6 Mice

Test Article: 3114 - SC

Mouse/C57BL/6

		Females				
		Treatment				
Group #	Category, Observations	Day of Phase:	1	4	8	11
1	Skin, Abrasion, Dorsal	Number Examined:	8	8	8	8
		Number Normal:	8	7	8	8
			0	1	0	0
2		Number Examined:	8	8	8	8
		Number Normal:	8	8	8	8
3	Skin, Abrasion, Dorsal	Number Examined:	8	8	8	8
		Number Normal:	8	7	7	7
			0	1	1	1
4	Skin, Abrasion, Dorsal	Number Examined:	1	1	1	1
		Number Normal:	1	0	0	0
			0	1	1	1

Table 2
Body Weights

Body Weight Summary Report with Individual Values (g)

Covance

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PATH/TOX SYSTEM Version 1.6.0 Build 27

Study: 8300180

StudyTitle: A Study to Assess Wound Healing in the C57BL/6 Mice

Test Article: 3114 - SC

Mouse/C57BL/6

Females

Group #	Animal #	Pretreatment Day: 0 Session 1n	Treatment Day: 7 Session 1n	Day: 13 Session 1n
1	1	16.5	18.2	18.4
	2	17.2	21.0	21.9
	3	19.6	19.6	20.0
	4	18.4	19.7	19.7
	5	19.3	19.7	19.9
	6	20.1	21.0	21.3
	7	16.9	18.2	18.5
	8	20.9	22.2	22.0
	(n)	8	8	8
	Means	18.61	19.95	20.21
	SDevs	1.62	1.40	1.41
2	9	19.9	20.3	20.5
	10	16.5	18.6	18.0
	11	18.0	18.7	18.7
	12	21.6	24.0	23.8
	13	16.8	18.7	18.4
	14	19.3	20.3	20.2

n= Analysis of Variance not significant. No
Unplanned test was performed

Body Weight Summary Report with Individual Values (g)

Covance

Printed: 16 Sep 2014 09:50:31 AM
 PATH/TOX SYSTEM Version 1.6.0 Build 27

Study: 8300180

StudyTitle: A Study to Assess Wound Healing in the C57BL/6 Mice

Test Article: 3114 - SC

Mouse/C57BL/6

Females				
Group #	Animal #	Pretreatment Day: 0 Session 1n	Treatment Day: 7 Session 1n	Day: 13 Session 1n
2	15	17.4	18.3	19.1
	16	20.4	21.9	22.6
	(n)	8	8	8
	Means	18.74	20.10	20.16
	SDevs	1.84	2.00	2.08
3	17	17.0	17.6	17.8
	18	20.8	21.9	22.3
	19	14.0	20.5	20.3
	20	18.6	19.4	19.6
	21	19.4	20.6	21.0
	22	19.9	21.2	21.3
	23	17.6	19.5	19.5
	24	21.8	20.5	20.9
	(n)	8	8	8
	Means	18.64	20.15	20.34
	SDevs	2.45	1.31	1.37
4	25	12.6	17.6	19.2

n= Analysis of Variance not significant. No
 Unplanned test was performed

Body Weight Summary Report with Individual Values (g)

Covance

Printed: 16 Sep 2014 09:50:31 AM

PATH/TOX SYSTEM Version 1.6.0 Build 27

Study: 8300180

StudyTitle: A Study to Assess Wound Healing in the C57BL/6 Mice

Test Article: 3114 - SC

Mouse/C57BL/6

Females				
Group #	Animal #	Pretreatment Day: 0 Session 1n	Treatment Day: 7 Session 1n	Day: 13 Session 1n
4	(n)	1	1	1
	Means	12.60	17.60	19.20
	SDevs	-	-	-

n= Analysis of Variance not significant. No

Unplanned test was performed

PROTOCOL AND AMENDMENTS



Study Protocol

A Study to Assess Wound Healing in the C57 BL/6 Mice

Covance Study Number: 8300180

Test Facility: Covance Laboratories Inc.
671 South Meridian Road
Greenfield, IN 46140

Building/Room: Building 241/927

Sponsor: Lpath
4025 Sorrento Valley Blvd
San Diego, CA 92121

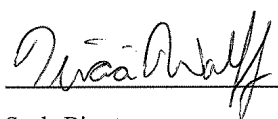
Planned Pre-Treatment Start Date: 15-Jul-2014

Planned Start Date: 16-Jul-2014

Planned End Date: 30-Jul-2014

Regulatory Status: Non-GLP

Protocol Approval:



Study Director

13 July 2014

Date

Sponsor Protocol Approval

Finalization of this protocol by the Study Director was authorized by the Sponsor via e-mail on 11-Jul-2014.

Key Personnel

Sponsor Representative	Gary Woodnutt Lpath Inc. 4025 Sorrento Valley Blvd San Diego, CA 92121 Telephone: 858-678-0800 ext. 127 Email: gwoodnutt@lpath.com
Study Director	Tricia Wolff Covance Laboratories Inc. Telephone: 317-467-7394 (desk), 317-405-7921 (mobile) E-mail: tricia.wolff@covance.com
Alt. Study Director	Susan Christopher Covance Laboratories Inc. Telephone: 317-467-2895 (desk), 317-966-6137 (mobile) E-mail: susan.christopher@covance.com
Scientific Lead	Subba R Chintalacharuvu (Chinne) Ph.D. Covance Laboratories Inc. Telephone: 317-467-7394 (desk) Email: subba.chinne@covance.com
Study Technician	Luke McGuire
Supervisor	Jason Anderson
Necropsy Technician	Zachary Prast
Pathologist	Pamela Blackshear
Surgical Technician	Marsha Longstreet

Shipping Details

Pharmacokinetic Samples (Plasma)	Gary Woodnutt Lpath Inc. 4025 Sorrento Valley Blvd San Diego, CA 92121 Telephone: 858-678-0800 Email: gwoodnutt@lpath.com
	Ship samples on dry ice Contact the recipient prior to shipment/ when shipped

Covance Study Number: 8300180

Study Identification

PHS* (Please note: all test articles with unknown hazards should be handled as PHS in neat (powder, etc) form. Once formulated, standard PPE will be used for handling of unknowns.)	No qualifying compounds
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*Particularly Hazardous Substance: Carcinogen, Reproductive, Acutely Toxic, or Highly Potent Materials. Additional PPE may be required. See the Test Material Safety Review (TMSR) for details. The TMSR is not considered raw data; additional PPE requirements will not be documented.

Subject Details

Treatment Grp	Dose (mg/kg)	Test Article	Animal Numbers	No. of Females
01	0	Vehicle	1-8	8
02	10	3114	9-16	8
03	50	3114	17-24	8
			Total	24

Grp. = Group; No. of Females = Number of Females

Covance ACUA Protocol	14-089
Species and Strain	Mouse-C57BL/6
Sex	Female
Source	Harlan
Vendor Nomenclature	C57BL/6NHsd
Approximate Age	9-10 weeks
Quantity to Order	26
Quantity Enrolled on Study	24
Distribution	Randomization will be based on body weight
Supplier/Food Type	Rodent Diet 2014, Teklad
Water	Greenfield- city water
Housing	Animals will be single housed in a Stainless-steel cages
Environmental Conditions	Standard Photoperiod: 12 hours light, 12 hours dark (may be interrupted for study-related activities) Temperature: 68-79°F Relative humidity: 30% – 70%

Covance Study Number: 8300180

Test Article and Dose Formulation Details

Test Article	Lot Number	Concentration (mg/mL)
3114	103948	32.1

Tx Grp	Test Article	Dose (mg/kg)	Dose Volume (mL/kg)	Concentration (mL/kg)	Dosing Frequency	Prep Frequency	Route
01	Vehicle	0	5	0	Twice weekly	Daily	SC
02	3114	10	1	10	Twice weekly	Daily	SC
03	3114	50	5	10	Twice weekly	Daily	SC

Abbreviations: Tx Grp = Treatment group; SC = Subcutaneous injection

Dosing Days	Day 1, 4, 8 and 11
Test Article	3114 Lot # 103948
Vehicle	Formulation buffer
Frequency of Preparation for Test Article	Daily
Correct for Potency	No
Average Animal Body Weight Factor	40 grams
Dose Volume Adjustment	Calculate doses based on most recent body weight.
Test Article Storage Conditions	Store at 4°C- Do not Freeze and Thaw
Test Article Mix Instructions	Dilute the stock concentration with the appropriate amount of formulation buffer. Swirl gently to mix.
Dose Preparation Storage Conditions	Store at 4°C
Disposition of Dose Preparations	Dispose of after expiration date or last scheduled dose.
Test Article Return Shipment	Any remaining test articles and excipient can be disposed of following the completion of the live phase of the study.

Live Phase Parameters***Pretreatment Parameters***

Randomization	Body weigh stratification on Day -1
Acclimation	Housed in vivarium at least 1 week prior to study start

Covance Study Number: 8300180

Clinical Signs	Daily for mortality
Body weight	Pre-treatment weight will be collected for randomization on Day -1

Surgery

Pre-Surgery Preparation	<u>Surgical Preparation</u> One or two days prior to surgery, all animals will be shaved and a depilatory (Nair®) will be applied to the surgical area to remove the hair. The animals will be anesthetized to effect using 1% to 5% Isoflurane gas in oxygen. Eye lubricant will be placed in the animals eyes to prevent drying of the corneas during anesthesia. The mice will be dosed with 0.05 mg/kg Buprenorphine IM/SC. The skin at the wound sites will be scrubbed with appropriate antiseptic (i.e. betadine, chlorhexidine) and wiped with sterile saline or isopropyl alcohol solution. Body temperature will be maintained via water heating pad or disposable thermal pad. Surgical instructions are previously sterilized and aseptic technique will be used throughout the procedure.
Surgery	Two wounds will be made as follows: Cranial wound: On the level of the dorsal midline at the scapula/thoracic area, a sterile 6-mm punch biopsy tool used to cut through the epidermis and dermis, and then scissors will be used to detach the tissue. Full thickness wounds extending through the panniculus carnosus will be made using the scissors. Caudal wound: Approximately 1 cm caudal to the cranial wound, a 1-cm linear incision will be made through the skin and subcutis along the dorsal midline. After photography at surgery, 20 µL of saline, will be administered directly to the wound for each group. And a new Opsite membrane will be adhered to both wounds. Tincture Benzoin compound will be applied outside of the perimeter of the wounds and the semi permeable polyurethane dressing OpSite will be placed over the wound and sealed at the edges by the Benzoin.
Post Operative Care	Following the procedures, the animals will be transferred to a warm recovery cage and monitored until fully recovered. Animals will then be returned to the home cage and will be checked at least once daily after the surgical procedure is performed. Signs of pain, swelling, infection or other abnormalities will be monitored.

Treatment Parameters

Clinical Observations	<u>Dosing Days:</u> Detailed observations will be done on each dosing day prior to each dose and approximately 1 hour post each dose. <u>Surgery Day:</u> See preceding section 'Post Operative Care' (under the section on 'Surgical Skin Incision'). <u>Note:</u> Additional clinical observations may be collected to characterize clinical changes.
Dose Frequency	Twice weekly; Days 1, 4, 8 and 11
Body weights	Days 7 and 14
<u>Data Collection</u>	<u>Dosing, Clinical Observations and Body Weights:</u> On-line (may be collected off-line on surgery days, but will be back-entered into Xybion). <u>All other Data:</u> Off-line
Animal Disposition	Euthanize and discard each animals after its necropsy

Wound Diameter and Photography

Photography	Days 1, 3, 6, 10, 13, and terminal. Collect photographs of the cranial and caudal wounds from each animal. Ensure a scale measurement is included with each photograph next to the wound and the same camera height is used for all photographs. Animals will be anesthetized to effect using 1% to 5% isoflurane gas in oxygen before collecting OpSite pouch removal
Wound Diameter	Determine the wound diameter from using Image J. Wound diameter should be measured based on the photography schedule, above.
Wound Site Coverage	After photography at surgery, 20 µL of saline will be administered directly to the wounds and a new OpSite membrane will be adhered to both wounds. Tincture Benzoin compound will be applied outside the perimeter of the wounds and the semi permeable polyurethane dressing OPSite will be placed over the wound and sealed at the edges by the Benzoin. The membrane will be removed for photographic procedures. After these procedures, saline and a new membrane will be applied.

*Wound Tensile Strength Assessment is described in the section of this protocol following Anatomic Pathology.

Toxicokinetic Animals Collection

Collect blood as described below from all available animals.

Specimen Number	Specimen Type	Collected By	Study Days	Time Point	Tube Prep and Processing	Collection Site/Method	Sample Volume	Anticoagulant and/or Tube Type	Pre-processing Storage	Centrifuge to Obtain
1	Blood	Necropsy	Necropsy	NA	SPAM	Cardiac Puncture	As much as possible	K ₃ -EDTA	Wet ice	Plasma
Special Instructions:										

Abbreviations: SPAM = Specimen Processing and Management.

Pharmacokinetic Animals Aliquots and Processing

Process samples from animals as described below.

Specimen Number	Aliquot Number	Container Prep	Sample Container Type	Sample Size	Destination	Assay(s)	Processing/Storage Conditions
1	1	SPAM	Screw top tubes	All – net volume	Sponsor	PK	Centrifuge to obtain plasma. Store frozen at ≤ -60°C until shipment.
Special Instructions		Ship to address in Shipping Details at completion of study.					

Abbreviations: SPAM = Specimen Processing and Management; PK = Pharmacokinetics.

Euthanasia

Euthanize animals by carbon dioxide (or anesthesia) followed by cervical dislocation, decapitation, bilateral thoracotomy, exsanguination, or vital organ removal to ensure death. Any animal that does not survive to study termination will be processed as per protocol. Tissues will be processed and evaluated at the request of the Study Director.

Anatomic Pathology**Necropsy**

Collect skin from the animals on Day 15 as described below.

Remove a rectangular section of skin, including the subcutis that contains the wound and extends approximately 2 cm lateral to the wound edges, taking care not to stretch or tear the wound. Lay flat on a non-stick surface and labeled to maintain orientation. (cranial/caudal). Carefully separate the 2 wounds sites. For the caudal wound, lay flat on a piece of thick paper with an arrow to mark cranial orientation, and place in 10% neutral buffered formalin for histology. For the cranial wound, assess for wound tensile strength described below. Collect a strip, approximately 0.5 cm in length (cranial to caudal across the widest portion of the wound for tensile strength assessment).

Formalin-fixed tissues will be stored until it is determined (by the Sponsor) it will be examined histopathologically. The outcome of the decision will be added by an amendment to this protocol.

Wound Tensile Strength

Uniform sections of the cranial skin wound will be obtained from each animal at necropsy. The skin section will be collected into 0.9% saline and positioned individually in the tensiometer grips. The tensiometer will then apply stretching force at a fixed rate to the skin until sample failure (the point the skin is torn). The instrument will report the maximum force that caused the sample to failure. After failure, the remaining pieces will be discarded.

Statistics and Reporting

Statistical analysis and interpretation of the results is the responsibility of the sponsor.

Animal Care and Use Statement

All procedures in this protocol are in compliance with the U.S. Department of Agriculture's (USDA) Animal Welfare Act (9 CFR Parts 1, 2, and 3); the Guide for the Care and Use of Laboratory Animals (Institute for Laboratory Animal Research, The National Academies Press, Washington, D.C.); and the National Institutes of Health, Office of Laboratory Animal Welfare (for NIH funded studies). Whenever possible, procedures in this study are designed to avoid or minimize discomfort, distress, and pain to animals.

Alteration of Design

Prior to making changes to the approved protocol, the Study Director or Alternate will consult the Sponsor as appropriate. If applicable, the date that Sponsor approval is received will be documented in the amendment.

Regulatory Compliance

This non-clinical laboratory study is not intended to be conducted in full accordance with the United States Food and Drug Administration (FDA) Good Laboratory Practice (GLP) regulations, 21 CFR Part 58, but will be conducted in accordance with Covance standard operating procedures.

Veterinary Care/Treatment

This study will comply with all applicable sections of the Guide for the Care and Use of Laboratory Animals. Whenever possible, procedures used in this study are designed to avoid or minimize discomfort, distress, and pain to animals. All procedures are described in this study protocol or in written laboratory procedures. These procedures are based on the most current available technologies concerning proper laboratory animal use and management.

Palliative and Prophylactic Measures

Following study distribution, the Study Director and Sponsor/Designee (if possible) will be included in discussions of palliative and prophylactic procedures (nonlife-threatening conditions, including suspension of dosing and removal of animals from study) recommended by the attending veterinarian. Final authority for decision making will be with the Laboratory Animal Veterinarian.

Reports

Final Report

An electronic copy of the final report will be available to the Sponsor.

Record Retention

The raw data, documentation, histology specimens, protocol, and final report for this study will be stored in the Covance archives for at least 1 year after report finalization. All other specimens/samples will be discarded appropriately unless otherwise specified in the protocol. The Covance archives staff will contact the Sponsor after 1 year to determine disposition of the archived materials. Raw data on durable media and copies of the protocol and final study report will be retained in the Covance Archives.

The Sponsor will be responsible for the maintenance of the test and control article reserve samples. Raw data, documentation, records, specimens, and contributor reports generated by the Sponsor as a result of this study will be archived in the storage facilities of the Sponsor.

Estimated Study Calendar

Date	Day Number	Task Name
09-Jul-2014	-7	Rec Animals
15-Jul-2014	-1	PT-BW
15-Jul-2014	-1	PT Dist to Grps
16-Jul-2014	1	Dose Admin.
16-Jul-2014	1	Photography
18-Jul-2014	3	Photography
19-Jul-2014	4	Dose Admin.
21-Jul-2014	6	Photography
22-Jul-2014	7	BW
23-Jul-2014	8	Dose Admin.
25-Jul-2014	10	Photography
26-Jul-2014	11	Dose Admin.
28-Jul-2014	13	Photography
29-Jul-2014	14	BW
30-Jul-2014	15	Photography
30-Jul-2014	15	Nec



Protocol Amendment 1

Study Number: 8300180

This amendment modifies the following portions of the protocol.

Effective Date: 16-July-2014

1. **Subject Details.** Replace the table in this section with the following.

Treatment Grp	Dose (mg/kg)	Test Article	Animal Numbers	No. of Females
01	0	Vehicle	1-8	8
02	10	3114	9-16	8
03	50	3114	17-24	8
04	10	3114	25	1
			Total	25

Grp. = Group; No. of Females = Number of Females

Reason for Change:

Replacement animal added to study due to an injury to a Group 02 animal during surgery.

2. **Subject Details, Quantity Enrolled on Study.** Replace the text in this section with the following.

Quantity Enrolled on Study	25
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Reason for Change:

Replacement animal added to study due to an injury to a Group 02 animal during surgery.



3. **Test Article and Dose Formulation Details.** Replace the tables in this section with the following.

Test Article	Lot Number	Concentration (mg/mL)
3114	GA121459	32.1

Tx Grp	Test Article	Dose (mg/kg)	Dose Volume (mL/kg)	Concentration (mL/kg)	Dosing Frequency	Prep Frequency	Route
01	Vehicle	0	5	0	Twice weekly	Daily	SC
02	3114	10	1	10	Twice weekly	Daily	SC
03	3114	50	5	10	Twice weekly	Daily	SC
04	3114	10	1	10	Twice weekly	Daily	SC

Reason for Change:

To correct the lot number of the test article, and to add Group 4 due to the addition of the replacement animal to the study.

4. **Test Article and Dose Formulation Details, Test Article.** Replace the text in this section with the following.

Test Article	3114 Lot # GA121459
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Reason for Change:

To correct the lot number of the test article

5. **Wound Diameter and Photography, Photography, and Wound Site Coverage.** Add the following to this section.

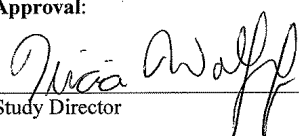
Photographs will be taken on Day 8, after photography saline will be applied to the wounds but a new OpSite membrane will not be adhered.

Reason for Change:

An extra day of photography is requested to monitor the wound healing process.

Covance 8300180
Protocol Amendment 1

Approval:


Study Director

23 Jul 2014
Date



Protocol Amendment 2

Study Number: 8300180

This amendment modifies the following portions of the protocol.

Effective Date: 25-July-2014

1. **Planned End Date.** Replace this section with the following

29-Jul-2014

Reason for Change:

Due to the progression of the wound healing the study will be terminated early.

2. **Treatment Parameters, Body Weights.** Replace this section with the following.

Body weights	Days 7 and 13
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Reason for Change:

Due to the study being terminated early the body weights are moved to Day 13.

3. **Wound Diameter and Photography, Photography and Wound Site Coverage.**
Replace this section with the following.

Photography	Days 1, 3, 6, 9, 13, and terminal. Collect photographs of the cranial and caudal wounds from each animal. Ensure a scale measurement is included with each photograph next to the wound and the same camera height is used for all photographs. Animals will be anesthetized to effect using 1% to 5% isoflurane gas in oxygen before collecting OpSite pouch removal
Wound Site Coverage	After photography at surgery, 20 μL of saline will be administered directly to the wounds



Reason for Change:

To correct the Day of photography from Day 8 to Day 9, and to remove the requirement to place a new Opsite pouch after photography.

4. **Anatomic Pathology, Necrospy.** Replace this section with the following.

Collect skin from the animals on Day **14** as described below.

Remove a rectangular section of skin, including the subcutis that contains the wound and extends approximately 2 cm lateral to the wound edges, taking care not to stretch or tear the wound. Lay flat on a non-stick surface and labeled to maintain orientation. (cranial/caudal). Carefully separate the 2 wounds sites. For the cranial wound, assess for wound tensile strength described below. Collect a strip, approximately 0.5 cm in length (cranial to caudal across the widest portion of the wound for tensile strength assessment.

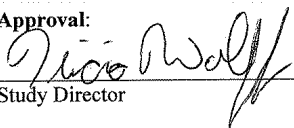
The caudal wound will be discarded without further evaluation.

Reason for Change:

Due to the progression of the wound healing the study will be terminated early and the caudal wound will not be saved for possible histopathology.

Covance 8300180
Protocol Amendment 2

Approval:


Study Director

25 July 2014
Date

CONTIBUTOR REPORTS

Pharmacokinetic Report

Human IgG Quantification ELISA

LTLT3114 antibody plasma concentrations were quantified using a direct-binding immunoassay (ELISA) method. Goat-anti human IgG (gamma-chain specific) UNLB antibody (Southern Biotech 2040-01, 1mg/mL) was diluted 1:1000 in carbonate buffer (100mM NaHCO₃, 33.6 mM Na₂CO₃, pH 9.5). 384 well plates were coated with 15µL/well of this coating solution and incubated at 4°C overnight. The plates were then washed 4 times with PBS and blocked with 50 µL/well blocking buffer (1%BSA, 0.1% Tween-20 in PBS) for 1 hour at room temperature. Samples and standard were prepared on non-binding plates with enough volume to run in duplicate. The standard was prepared by diluting human anti-LPA antibody LTLT3114 in blocking buffer (1%BSA, 0.1% Tween-20 in PBS) to the following dilutions: 900 ng/ml, 300 ng/ml, 100 ng/ml, 33.33 ng/ml, 11.11 ng/ml, 3.70 ng/ml, 1.23 ng/ml, 0.41 ng/ml, 0.14 ng/mL, 0.05 ng/mL, 0.02 ng/mL, and 0 ng/mL. The samples were diluted in blocking buffer 1:10, 1:30, 1:90, 1:270, 1:810, 1:2430, 1: 7290, 1: 21870, 1: 65610, 1: 196380 and 1: 590490 and 15µL of the standard/samples preparation were loaded to each well and incubated at room temperature for 1 hour. Plates were washed 4 times with PBS and 15 µL of 1: 10,000 blocking buffer diluted goat anti-human IgG HRP (Southern Biotech 2040-05) was added to each well and incubated for 1 hour at room temperature. Then, the plates were washed 4 times with PBS and developed using 15µl/well TMB substrate at 4°C. After 5-20 minutes, the reaction was stopped with 1M H₂SO₄, 15µl/well. The OD was measured at 450 nm. Data was analyzed using Graphpad Prism software. The standard ODs were graphed using a four parameter equation and this was used to calculate the human IgG in the samples. Excel software was used to calculate the µg/ml of human IgG in each sample by correcting the values for the dilution factor.

Antigen Laid-Down ELISA

384 well plates were coated with 15 µl of 1 µg/ml of (C12) LPA-BSA-SMCC diluted in carbonate buffer (100mM NaHCO₃, 33.6 mM Na₂CO₃, pH 9.5) and incubated at 4°C overnight. The plates were washed 4 times with PBS and blocked with 30µl of blocking buffer (1%BSA, 0.1% Tween-20 in PBS) for 1 hour at room temperature. 15µl of LTLT3114 antibody was added starting at 4µg/ml and diluted 2-fold in blocking buffer across the plate for the standard curve. Rat plasma sample were diluted to 1:1000, 1:2000, 1:4000, 1:8000, 1:16000 and 1:32000 in blocking buffer, and 15µl of each dilution was ran in duplicate. Plates were incubated at room temperature for 1 hour. Next, plates were washed 4 times with PBS and 15µl of a peroxidase conjugated anti-human IgG (H+L) (Jackson immunoresearch 709-035-149) was added at a 1:40K dilution and incubated for 1 hour at room temperature. Finally, the plates were developed with 15µl

cold TMB substrate for 12 minutes at room temperature. The reaction was stopped with 15µl of 1M H₂SO₄. Plates were read at 450 nm.

RESULTS

LTLT3114 measurement in plasma samples (GA121459)

			[LT3114] µg/mL	
Label	Group	Treatment	IgG Quant	PK ELISA
8300180-1-F-grp1-wound.JPG	1	vehicle	0	0
8300180-2-F-grp1-wound.JPG			0	0
8300180-3-F-grp1-wound.JPG			0	0
8300180-4-F-grp1-wound.JPG			0	0
8300180-5-F-grp1-wound.JPG			0	0
8300180-6-F-grp1-wound.JPG			0	0
8300180-7-F-grp1-wound.JPG			0	0
8300180-8-F-grp1-wound.JPG			0	0
8300180-9-F-grp2-wound.JPG	2	Low Dose 10mg/kg	194.4	103.7
8300180-10-F-grp2-wound.JPG			239.4	161.9
8300180-11-F-grp2-wound.JPG			239.6	100.6
8300180-12-F-grp2-wound.JPG			130.4	117.2
8300180-13-F-grp2-wound.JPG			167.4	118.1
8300180-14-F-grp2-wound.JPG			184.5	197.6
8300180-15-F-grp2-wound.JPG			212.8	120.7
8300180-16-F-grp2-wound.JPG			110.1	87.1

wound.JPG				
average			184.8	125.9
8300180-17-F-grp3-wound.JPG	3	High Dose 50mg/kg	395.9	139.4
8300180-18-F-grp3-wound.JPG			169.3	129.2
8300180-19-F-grp3-wound.JPG			449.2	247.2
8300180-20-F-grp3-wound.JPG			340.0	223.1
8300180-21-F-grp3-wound.JPG			208.1	186.4
8300180-22-F-grp3-wound.JPG			183.8	222.4
8300180-23-F-grp3-wound.JPG			172.8	131.9
8300180-24-F-grp3-wound.JPG			217.7	224.0
average			267.1	188.0
8300180-25-F-grp4-wound.JPG	4		136.3	123.0

DATA LOCATION

Notebook 316, Page 44-47, 65-72. (Protocol is on Page 44-47 and Page 67, plate layout is on Page 68, plate Macro data is on Page 69-71, result is on Page 72)

Electronic files: L-Drive Location: \\WSIG-SRV01\Lpath\WORKING FOLDERS-LPATH\Ying\2014\PK\10-2014 PK ELISA Wound healing\10-1-14 wound healing